Rishi Rakesh Agrawal

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech in Mechanical Engineering	Indian Institute of Technology Kanpur	8.44/10
2021	Class XII (GSEB)	Green Valley Higher Secondary School	91.38%
2019	Class X (GSEB)	Gujarat Public School	84.84%

Scholastic Achievements

- Received **OPJEMS Award** in academic year **2022-23** for being in top 100 students across 45+ Premier Institutes
- Awarded **Academic Excellence Award** twice in academic years **2021** and **2022** for exceptional performance in academics
- Achieved **AIR 2566** in **JEE Advanced 2021** and **AIR 2980** in **JEE Mains 2021** among 1.2+ million candidates

Work Experience

Student Research Associate || **Industry 4.0** || Prof. Nalinaksh S. Vyas *(IIT Kanpur)* | *May'23 - Present*

Objective	• Development of Indigenous **Industry 4.0** Software and Hardware Platform for Ordnance Production
Approach	• Implemented & Integrated **Enterprise Resource Planning** System to automate **350+** machine operations • Enhanced android app's **REST API** in **NodeJS** for real-time tracking of employee data, location by admins • Implemented **Real Time-Machine Monitoring** by obtaining data through **NI-OPC** servers on **LabView**
Impact	• Improved overall **Productivity** by **17%**, **Diagnosis**, **maintainability** along **Dynamic** process planning

Full Stack Unity Game Developer || *Vizuara* | *Apr'23 - Jun'23*

Objective	• Development of high quality **interactive** digital textbooks at the intersection of Education and **Gamification**
Approach	• Developed robust, high-quality & engaging games utilizing **Unity** & **Blender** that made learning interactive • Overseen end-to-end game development process, encompassing **Programming**, **3D Art**, **UI/UX**, **Quality**
Impact	• **Elevated educational experience** by delivering cutting edge 3D, gamified & interactive contents to schools

Key Projects

Project Mentor: Game Anti-Cheat Client || *Game Development Club, IIT Kanpur* *May'23 - Jul'23*

Mentor	• Mentored **10 freshers** by guiding on fundamentals of **Anti-Cheat**, led development of **Cheat Detection**
Approach	• Researched on various hacking techniques, including **AimBots** and **WallHacks** and existing hacking software • Developed and Implemented a robust and effective **Server Side** Anti Cheat system for multiplayer games

Mathematical foundations of Computer Graphics || *Stamatics, IIT Kanpur* *May'23 - Jul'23*

Objective	• Gain deep understanding on principals of Computer Graphics like **Ray Tracing** & **Procedural Generation**
Approach	• Implemented Ray Tracing Algorithm in **C++**, incorporating **Blinn Phong** shading model for lighting effect • Implemented Procedural Generation techniques like **Perlin Noise** developing natural looking visual elements

Option and Strategies to Hedge a Corpus || *Finance and Analytics Club, IIT Kanpur* *May'22 - Jul'22*

Objective	• Evaluate **16+ Option Strategies** to efficiently **Hedge a Corpus** against market risk and fluctuations
Approach	• Calculated & analysed **Greek Function** & conducted comprehensive **Risk Assessment** for 4 option strategy • Utilized **Opstra** to implement **backtests**, running multiple simulation to evaluate strategy performance

Positions of Responsibility

Coordinator, Game Development Club IIT Kanpur *Apr'23 - Present*

Leadership	• Spearheaded a **3-tier team** of **30+** members to organise events, workshops, sessions for campus community • Coordinated with Core Team to organize Semester Projects, for **100+ students** on topics Anti-Cheat & AI
Initiatives	• Delivered **influential** workshops on GameDev, attracting **100+** motivated developers enhancing their skills
Impact	• Received a **funding** of **INR 1 Lakh+** from **Vizuara** as a result of successful **Industrial Collaboration**

Web Organizer, Udghosh IIT Kanpur *Jul'22 - Oct'22*

Teamwork	• Collaborated with a team of **10+** to gather requirements, design and develop the official website of Udghosh
Initiatives	• **Analyzed** website's performance & made data-driven changes to enhance its functionality & user experience
Impact	• **Strategically** elevated festival budget by **INR 45 Lakh** though website by attracting sponsorship & funding

Technical Skills

Programming: C, C++, JavaScript, Python, C#
Development: ReactJS, NodeJS, HTML, CSS
Utilities: Git, NumPy, Pandas, MongoDB, MySQL
Softwares: MS Office, VS Code, Unity

Relevant Coursework

Design and Analysis of Algorithms	Data Structures and Algorithms	Fundamentals of Programming
Philosophical Logic	Introduction to Critical Thinking	Introduction to Economics
Introduction to Electronics	Partial Differential Equation	Linear Algebra

Extra Curricular Activities

Technical	• Achieved **Rank 1** in **SnT Code**, a 10+ hour Inter Hall Competition demonstrating Problem Solving skills • Secured **Rank 1** representing my Hall in **Takneek'22**, the prestigious Inter Hall Technical Tournament • Participated in Problem Statement of The **IGDC GameDev Challenge** in **Inter IIT Tech Meet 11.0**
Sports	• Secured **Gold** representing IIT Kanpur **Hockey Team** in **Udghosh'22**, Annual Sports Festival of IITK • Participated in **Inferno'22**, the Inter Hall Sports Tournament representing my Hall on **Hockey ground**